

**2018 JC2 Preliminary Examination**  
**H1 Chemistry 8873**  
**Paper 1 Worked Solution**

- 1 Let the % of  $^{12}\text{C}$  by  $x\%$ .

So there is  $(100 - x)\%$  of  $^{13}\text{C}$ .

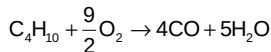
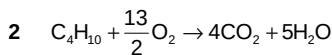
$$\frac{12x + 13(100 - x)}{100} = 12.20$$

$$1300 - x = 1220$$

$$x = 80$$

$$\therefore ^{12}\text{C}:^{13}\text{C} = 80 : 20 = 4 : 1$$

$\Rightarrow$  A



Since the same volume of  $\text{CO}$  and  $\text{CO}_2$  is produced, we have  $n_{\text{CO}} = n_{\text{CO}_2}$ :



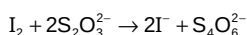
$$\text{Volume of O}_2 = \frac{11}{2} \times 30 = 165 \text{ cm}^3$$

$\Rightarrow$  D

3 O.S. of Mo =  $+\frac{-8 - (112 \times -2)}{36} = +6$

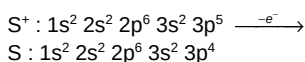
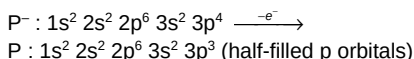
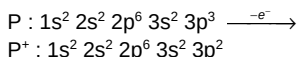
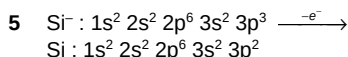
$\Rightarrow$  D

4  $n_{\text{I}_2} = \frac{1}{5}n_{\text{CO}} = \frac{1}{5} \times 1 = 0.2 \text{ mol}$



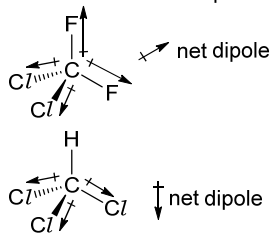
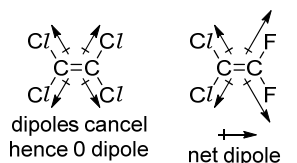
$$n_{\text{S}_2\text{O}_3^{2-}} = 2n_{\text{I}_2} = 2 \times 0.2 = 0.4 \text{ mol}$$

$\Rightarrow$  B



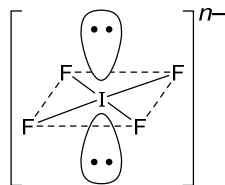
$\Rightarrow$  C

- 6 Bond dipoles are vectors which add up vectorially to give the resultant dipole of the molecules:



$\Rightarrow$  A

- 7 In order for  $\text{IF}_4^{n-}$  to adopt a square planar shape, there are two lone pairs in the structure:



Total number of electrons around I = 12  
of which 4 originates from the four F atoms (contributing to the 4 I-F bonds)  
and 7 from the I (Group 17) itself

Hence,  $n = 12 - 4 - 7 = 1$  (i.e. the iodine had gained one electron from elsewhere, thus a  $-1$  charge)

$\Rightarrow$  A

8 Heat evolved,  $q = mc\Delta T$   
 $= 250 \times 4.20 \times 15$   
 $= 15750 \text{ J}$

$$n_{\text{Zn}} = \frac{13.08}{65.4} = 0.200 \text{ mol}$$

$$n_{\text{Cu}^{2+}} = \frac{250}{1000} \times 1.0 = 0.250 \text{ mol}$$

Zn is the limiting reagent

$$\text{Hence, } \Delta H = -\frac{q}{n_{\text{Zn}}} = -\frac{15750}{0.200}$$

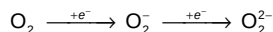
$$= -78750 \text{ J mol}^{-1}$$

$$= -78.8 \text{ kJ mol}^{-1}$$

$\Rightarrow$  C

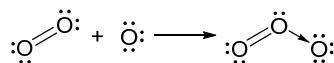
- 9 1 ✓: Involves breaking the  $\text{O}=\text{O}$  in  $\text{O}_2$  to give O atoms, hence endothermic

- 2 ✓: Electron affinity occurs in two stages:



The second electron affinity involves the addition of an electron to an already negative  $\text{O}_2^-$  anion, and would be highly endothermic, rendering the whole process endothermic

- 3 ✗: Involves the formation of a dative bond from  $\text{O}_2$  to O atom to give  $\text{O}_3$ , and hence exothermic:



$\Rightarrow$  B

10 lattice energy, L.E.  $\propto \left| \frac{q^+q^-}{r_+ + r_-} \right|$

Since  $r_{\text{Cs}^+} > r_{\text{Rb}^+}$  and  $r_{\text{Cl}^-} > r_{\text{F}^-}$ , so the numerical value of the lattice energy of  $\text{CsF}$  will be **in between** that of  $\text{CsCl}$  and  $\text{RbF}$

$\Rightarrow$  C

11  $K_c = \frac{[\text{H}_2]^4}{[\text{H}_2\text{O}]^4}$

$$\text{Units of } K_c = \frac{(\text{mol dm}^{-3})^4}{(\text{mol dm}^{-3})^4} = \text{dimensionless}$$

$\Rightarrow$  D

- 12 1 ✓:  $K_c$  changes with temperature. Hence if the students carry out the reaction at different temperatures, the values of  $K_c$  will differ

- 2 ✗:  $K_c$  is independent of the initial concentrations of reactants.  $K_c$  only depends on the temperature

- 3 ✗:  $K_c$  is independent of the amount of reactants and products.  $K_c$  only depends on the temperature

$\Rightarrow$  D

13  $n_{\text{H}_2\text{SO}_4} = \frac{20.0}{1000} \times 0.10 = 0.00200 \text{ mol}$

$$n_{\text{NaOH}} = \frac{20.0}{1000} \times 0.10 = 0.00200 \text{ mol}$$

Hence  $\text{H}_2\text{SO}_4$  is in excess

$$n_{\text{excess H}_2\text{SO}_4} = 0.00200 - \frac{1}{2} \times 0.00200$$

$$= 0.00100 \text{ mol}$$

$$[\text{H}^+] = \frac{2n_{\text{excess H}_2\text{SO}_4}}{\frac{20.0 + 20.0}{1000}} = \frac{0.00200}{40.0}$$

$$= 0.0500 \text{ mol dm}^{-3}$$

$$\text{pH} = -\lg[\text{H}^+] = -\lg(0.0500) = 1.30$$

$\Rightarrow$  B

- 14 If  $\text{H}^+$  are catalyst,  $[\text{H}^+]$  does not change as the reaction progresses, and hence a concentration-time plot (horizontal straight line) will not give information on order of reaction w.r.t.  $\text{H}^+$ .

Thus **B, C and D** which relies on a change in  $[\text{H}^+]$  with time does not work.

**A** which investigates the relationship between rate of reaction and  $[\text{H}^+]$  directly is the only valid method.

$\Rightarrow$  A

- 15 A ✗: Covalent character ↑es from  $\text{Na}_2\text{O}$  to  $\text{SiO}_2$ , but remains covalent from  $\text{SiO}_2$  to  $\text{P}_4\text{O}_{10}$

- B ✗: Melting point ↑es from  $\text{Na}_2\text{O}$  to  $\text{SiO}_2$ , but ↓es from  $\text{SiO}_2$  to  $\text{P}_4\text{O}_{10}$

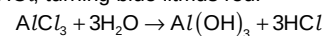
- C ✓: pH of solution of  $\text{Na}_2\text{O}$  (=14),  $\text{MgO}$  (=9),  $\text{Al}_2\text{O}_3$  and  $\text{SiO}_2$  (=7, insoluble),  $\text{P}_4\text{O}_{10}$  (=2)

- D ✗:  $\text{Na}_2\text{O}$  is soluble,  $\text{MgO}$  is sparingly soluble and  $\text{SiO}_2$  is insoluble in aqueous alkali (requires heating with concentrated  $\text{NaOH}$ ), but  $\text{P}_4\text{O}_{10}$  is soluble in aqueous alkali

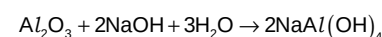
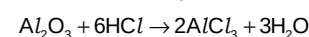
$\Rightarrow$  C

- 16 A ✓:  $\text{AlCl}_3$  is ionic in the solid state, becoming the covalent dimer only upon melting

- B ✗:  $\text{AlCl}_3$  undergoes extensive hydrolysis in water to give an acidic solution of  $\text{HCl}$ , turning blue litmus red:



- C ✗:  $\text{Al}_2\text{O}_3$  is amphoteric:



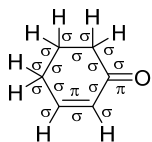
- D ✗:  $r_{\text{F}^-} < r_{\text{Br}^-}$ ,  $\therefore \text{L.E.} \propto \left| \frac{q^+q^-}{r_+ + r_-} \right|$  of  $\text{AlF}_3$  is more exothermic than that of  $\text{AlBr}_3$ , reflected in the higher melting point

$\Rightarrow$  A

- 17 **A ✗**:  $\text{Al}_2\text{O}_3$  is insoluble hence neutral (pH 7), while  $\text{AlCl}_3$  is acidic in water (pH 3)
- B ✗**:  $\text{MgO}$  is slightly soluble giving pH 9, while  $\text{MgCl}_2$  undergoes slight hydrolysis to give pH 6.5
- C ✓**:  $\text{P}_4\text{O}_{10}$  is acidic in water (pH = 2) and  $\text{PCl}_5$  is also acidic in water (pH = 2)
- D ✗**:  $\text{Na}_2\text{O}$  is alkaline in water (pH 14) while  $\text{NaCl}$  is neutral (pH 7)

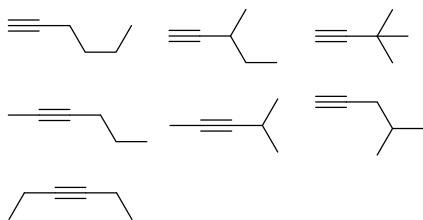
⇒ C

- 18 Structure of cyclohexanone, including H:



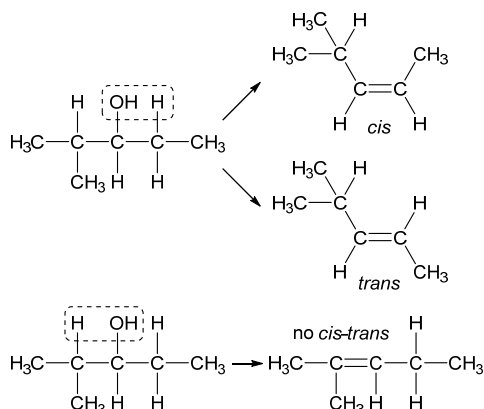
⇒ D

- 19 The structural isomers are



⇒ D

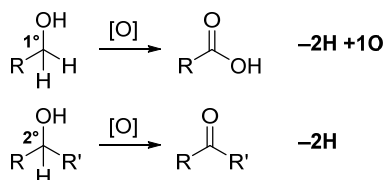
- 20 Elimination can occur with the H on both sides of the OH:



⇒ B

- 21  $\text{C}_4\text{H}_{10}\text{O}_2$  is saturated, hence is a diol, with 2 OH groups.

Oxidation with  $\text{KMnO}_4$  gives  $\text{C}_4\text{H}_6\text{O}_4$ , with loss of 4H and gain of 2O. Hence both alcohols are  $1^\circ$  alcohols:



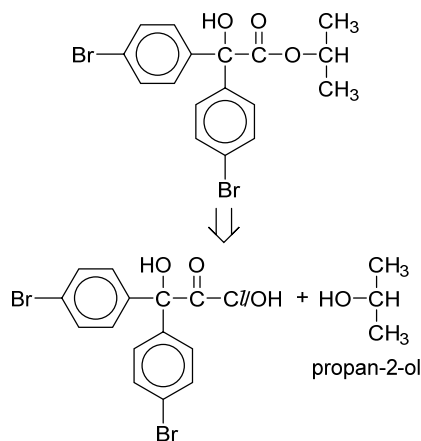
Since the X has an unbranched carbon chain, the OH groups must be on carbon 1 and 4.

⇒ C

- 22  $\text{CH}_3\text{CH}_2\text{CH}_2\text{O}-\text{C}(=\text{O})-\text{H}$  can only form  $\text{pd-pd}$  interaction and not H bond like in  $\text{RCO}_2\text{H}$ , hence lower b.p.
- propyl methanoate

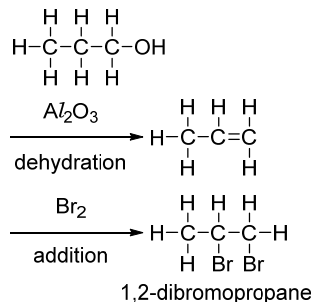
⇒ D

23



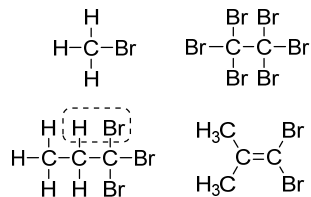
⇒ A

24



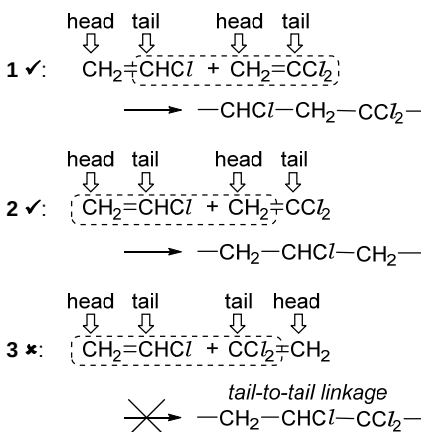
⇒ C

25



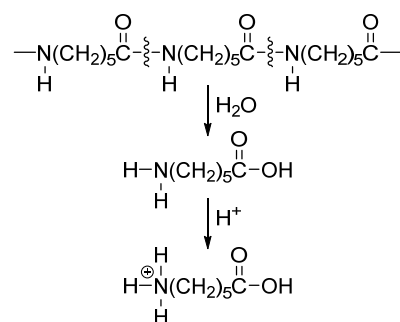
⇒ C

- 26 Only head-to-tail linkage is allowed.



⇒ A

- 27 Hydrolysis of the amide linkage:



⇒ D

- 28 **1 ✓**: PVC contains C, H and Cl, hence will form ( $\text{H}_2\text{O}$ ),  $\text{CO}$ ,  $\text{CO}_2$  and  $\text{HCl}$  upon combustion.

**2 ✗**: Poly(ethene), being a hydrocarbon, is essentially non-biodegradable.

**3 ✓**: An addition polymer is composed of  $n$  repeats of the monomer unit, hence will have the same empirical formula

⇒ B

29

$$n_{\text{glucose}} = [\text{glucose}] \times V_{\text{glucose}}$$

$$= (5 \times 10^{-4} \text{ mol dm}^{-3}) \times (4 \times 10^{-18} \text{ dm}^3)$$

$$= 2 \times 10^{-21} \text{ mol}$$

$$\text{No. of molecules} = n_{\text{glucose}} \times N_A$$

$$= (2 \times 10^{-21} \text{ mol}) \times (6.02 \times 10^{23} \text{ mol}^{-1})$$

$$= 1204$$

⇒ B

- 30 **1 ✗**: Use of graphene in tennis racquet is due to the strength of graphene sheets despite its low mass

**2 ✓**: Use of silver nanoparticles relies on its high surface area-to-volume ratio, which maximises the interaction between the pathogens and Ag

**3 ✓**: The efficiency of the catalyst depends on the surface available for the gaseous pollutants to bind to, hence the larger surface area-to-volume ratio will render the platinum nanoparticles more efficient

**4 ✓**: The colour of suspension of nanoparticles depends on the size of the nanoparticles due to surface plasmon resonance, involving excitation of the surface electrons. Hence changes in the surface area-to-volume ratio will affect the number of surface electrons, hence affecting the colour

⇒ B

### Answer Key

| Qn | Ans | Qn | Ans | Qn | Ans |
|----|-----|----|-----|----|-----|
| 1  | A   | 11 | D   | 21 | C   |
| 2  | D   | 12 | D   | 22 | D   |
| 3  | D   | 13 | B   | 23 | A   |
| 4  | B   | 14 | A   | 24 | C   |
| 5  | C   | 15 | C   | 25 | C   |
| 6  | A   | 16 | A   | 26 | A   |
| 7  | A   | 17 | C   | 27 | D   |
| 8  | C   | 18 | D   | 28 | B   |
| 9  | B   | 19 | D   | 29 | B   |

|    |   |    |   |    |   |
|----|---|----|---|----|---|
| 10 | C | 20 | B | 30 | B |
|----|---|----|---|----|---|